

GERMAINE HOUNAKEY

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Work Authorization

U.S. Citizen | Authorized to work in the U.S. | Available for immediate start.

EDUCATION

University of Michigan, Dearborn, MI

Master's in Computer Science - Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, HTML, CSS, SQL.

Operating Systems: UNIX, Linux, Windows.

Other Tools: Docker, Kubernetes, PyTorch, Git.

EXPERIENCE

Machine Learning Engineer | U.S. Army Corps of Engineers, ERDC-CRREL

Hanover, NH (Remote) • 01/2025 – 04/2026 • Hours per week: 40

- Built multi-modal object detection models in Python and OpenCV across RGB, thermal, and polarimetric LWIR imagery for defense research applications.
- Enabled reliable target detection in low-visibility and adverse-weather conditions by combining complementary thermal and polarimetric signals.
- Engineered image-processing pipelines that fused environmental and sensor data, improving object detection accuracy by 12% across diverse terrain and climate conditions.
- Designed data-collection and preprocessing workflows for thermal imaging hardware, processing thousands of images across multiple sensor modalities for downstream model training.
- Authored technical documentation and presented research findings to technical and non-technical stakeholders across cross-functional research and engineering teams.

Software Engineer | Corewell Health (Priority Health)

Grand Rapids, MI • 05/2022 – 07/2024 • Hours per week: 40

- Built and shipped responsive member portal features in Angular for Priority Health applications serving large-scale healthcare users.
- Developed and maintained ASP.NET backend services supporting core portal functionality and authentication workflows.
- Architected and implemented a biometric authentication workflow for the Priority Health iOS application, improving mobile login efficiency and user experience.
- Troubleshooted and resolved production issues across web and mobile platforms to maintain application reliability and performance.
- Collaborated with cross-functional teams to gather requirements, scope technical solutions, and deliver platform enhancements.

Software Engineer Intern | Corewell Health

Grand Rapids, MI • 04/2020 – 08/2020 • Hours per week: 40

- Contributed to feature development and testing within an Agile team, participating in standups and retrospectives.
- Authored test scenarios and tracked defects to improve software release quality.
- Produced technical documentation for IT systems and engineering processes.
- Drove process-improvement initiatives within the IT department.
- Supported system maintenance and operational tasks across a hybrid cloud and enterprise software environment.

PROJECTS

- [TasteCode](#), (Kotlin, Firebase, Android Studio) | March 2025
Built TasteCode, a modern Android recipe application using Kotlin, Jetpack Compose, Firebase Firestore, and Room Database, featuring real-time recipe search, offline storage, filtering, and user authentication workflows.
- [Intrusion Alert Explainer](#), (Python) | April 2026

Built Intrusion Alert Explainer, a Python-based cybersecurity tool that maps IDS and Snort alerts to CVE vulnerabilities using TF-IDF ranked retrieval, CVSS severity scoring, and CWE-based remediation guidance across 43,000+ NVD vulnerability entries.